

Who Shapes the Stance?

A Stases-Based Analysis of Writer-LLM Negotiation with CoAuthor

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An Argumentative Writing Stance

Writing is not just about the individual ideas or sentences. What distinguishes good writing, is ***how the ideas are structured and rhetorically organize*** [1] [2].

Position

***what** the writer chooses to argue*

Framing

***how** the writer organizes that position*

Support

***how** the position is backed up (experience or facts)*

Stasis Theory

In classical rhetoric, argumentative writers **consistently deliberate** and move through five types of **rhetorical choices** called **stases** [3].



CONJECTURAL

Is it? Did it happen?

Facts, claims,
examples, evidence



QUALITY

*Of what significance
is it?*

Judges value or
importance



CAUSE

What caused it?

Reasons,
consequences,
relationships



TRANSLATIVE

*What action is
appropriate?*

Proposes solutions or
conclusions



DEFINITIVE

What is it?

Defines or clarifies a
stated idea

+ **NONSENSE** — added to label incoherent or irrelevant suggestions (i.e., LLM hallucination).

Most LLM-based writing tools...

-  1 use acceptance, a binary unit of measurement (i.e., true or false), for assessment.
-  2 miss *how* the idea fits into the writer's argument (i.e., framing).
-  3 require explicit prompts, which interrupts the flow state these tools are meant to support

CoAuthor Tool & Dataset

- The tool is a *human-AI collaborative writing dataset* of 63 writers across 1445 writing sessions
- The dataset includes interactions with the tool at a keystroke level
 - Pressing tab prompts the LLM for suggestions
- LLM Suggestions
 - the LLM provides *at most 5 suggestions at a time*
 - the author can select *at most 1 suggestion at a time*

Our Study

20

argumentative
writing sessions

16

unique authors

174

prompts to
the LLM

734

candidate
suggestions

During argumentative writing, how do authors *implicitly negotiate* with CoAuthor's suggestions?

3 Codes Per Suggestion

01

TIMING

When the LLM is prompted (i.e., start or middle of sentence)

02

CONTENT

What the LLM rhetorically contributed (i.e., stasis theory)

03

OUTCOME

How the LLM's suggestion was ultimately used (i.e., reject, accept, edit, reverse)

Suggestion Block: A State of Implicit Negotiation

... Some people find the news to be depressing because they always cover death and disaster. But there are many news sources available these days that cover a variety of topics. <TAB> start of sentence

translative

1. If you enjoy reading, instead of watching TV for news, you can also read newspapers or blogs.

cause

2. This includes sports, entertainment, trends and even recipes.

cause

3. There's the local news, national news, world news, music news, science and technology news, fashion and style news, celebrity gossip and much more

definitive

4. There's no argument that it's important to be informed, but the real question is what constitutes being informed and how best to keep

nonsense

5. Brian Linke

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... Some people find the news to be depressing because they always cover death and disaster. But there are many news sources available these days that cover a variety of topics. **This includes sports, entertainment, trends, and even recipes.**

Thematic Analysis Results*

Content	Reject		Accept		Edit		Reverse		Total
	Start	Middle	Start	Middle	Start	Middle	Start	Middle	
Conjectural	98 (41%)	94 (40%)	18 (8%)	7 (3%)	6 (3%)	12 (5%)	2 (1%)	0 (0%)	237 (100%)
Quality	121 (53%)	64 (28%)	23 (10%)	8 (4%)	3 (1%)	5 (2%)	3 (1%)	0 (0%)	227 (100%)
Cause	43 (40%)	45 (42%)	4 (4%)	8 (7%)	5 (5%)	1 (1%)	1 (1%)	0 (0%)	107 (100%)
Translative	60 (59%)	19 (19%)	15 (15%)	3 (3%)	1 (1%)	3 (3%)	0 (0%)	0 (0%)	101 (100%)
Nonsense	32 (71%)	12 (27%)	1 (2%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	45 (100%)
Definitive	5 (29%)	6 (35%)	2 (12%)	1 (6%)	1 (6%)	1 (6%)	1 (6%)	0 (0%)	17 (100%)
Total	359 (49%)	240 (33%)	63 (9%)	27 (4%)	16 (2%)	22 (3%)	7 (1%)	0 (0%)	734 (100%)

*Cohen's Kappa: 0.74

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63% of suggestions are Conjectural or Quality.
The LLM leads with claims and opinions.

RQ: During argumentative writing, how do authors implicitly negotiate with CoAuthor's suggestions?

LLM Temperature 0.2 → 0.9

- *Cause, Conjectural, Definitive, & Nonsense* **doubles** in frequency
- *Quality* is **halved**
- *Translative* stays the **same**

Content	Temp 0.2	Temp 0.9
Cause	18 (9%)	89 (17%)
Conjectural	41 (20%)	196 (37%)
Definitive	2 (1%)	15 (3%)
Nonsense	6 (3%)	39 (7%)
Quality	108 (52%)	119 (23%)
Translative	34 (16%)	67 (13%)
Total	209 (100%)	525 (100%)

Location → Content $p < .05$

- *Start: Translative, Quality, Nonsense*
- *Middle: Conjectural & Cause*

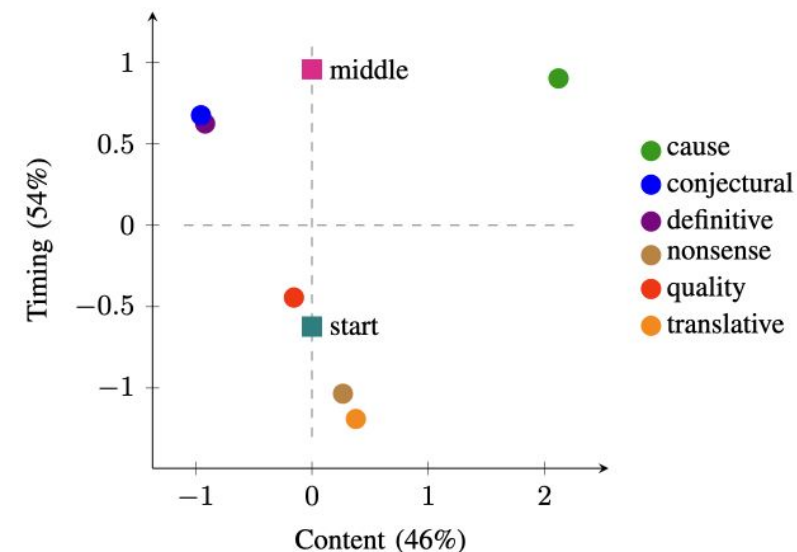
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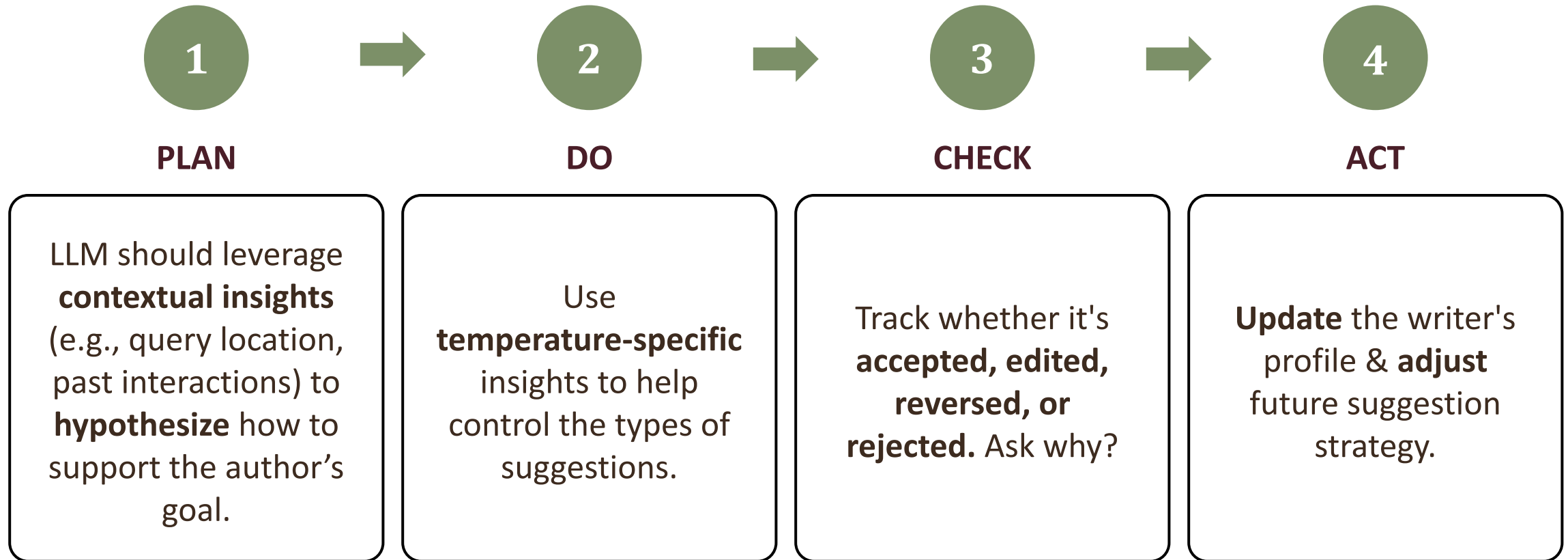
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Location → Content $p < .05$

- **Start:** *Translative, Quality, Nonsense*
- **Middle:** *Conjectural, Definitive, and Cause*



A writer-aware, iterative design using DevOp's PDCA



Rhetorical awareness and *contextual insights* should shape LLM suggestions.

- The LLM defaults to *leading the* argument rather than supporting the writer's own.
- LLM temperature and prompting location should be treated as a design characteristic for shaping the LLM's output.
- A DevOp's PDCA-style, writer-aware loop can align suggestions to each writer's rhetorical trajectory.

Thank you!

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Identify impacts outcome.

Multinomial Logistic Regression

- Content + Timing -> 4% variance on outcome
- + Writer -> +5% = 9%

TABLE II
MULTINOMIAL LOGISTIC REGRESSION MODEL COMPARISON.
**: SIGNIFICANT COMPARED TO M1

Model	Predictors	Log-Lik	McF. R ²
M0	Null (intercept only)	−436	.00
M1	Content + Timing	−424	.03
M2	Content * Timing	−417	.04
M3**	Content + Timing + Writer	−396	.09

Temperature impacts content.*

Content	Temp 0.2	Temp 0.9
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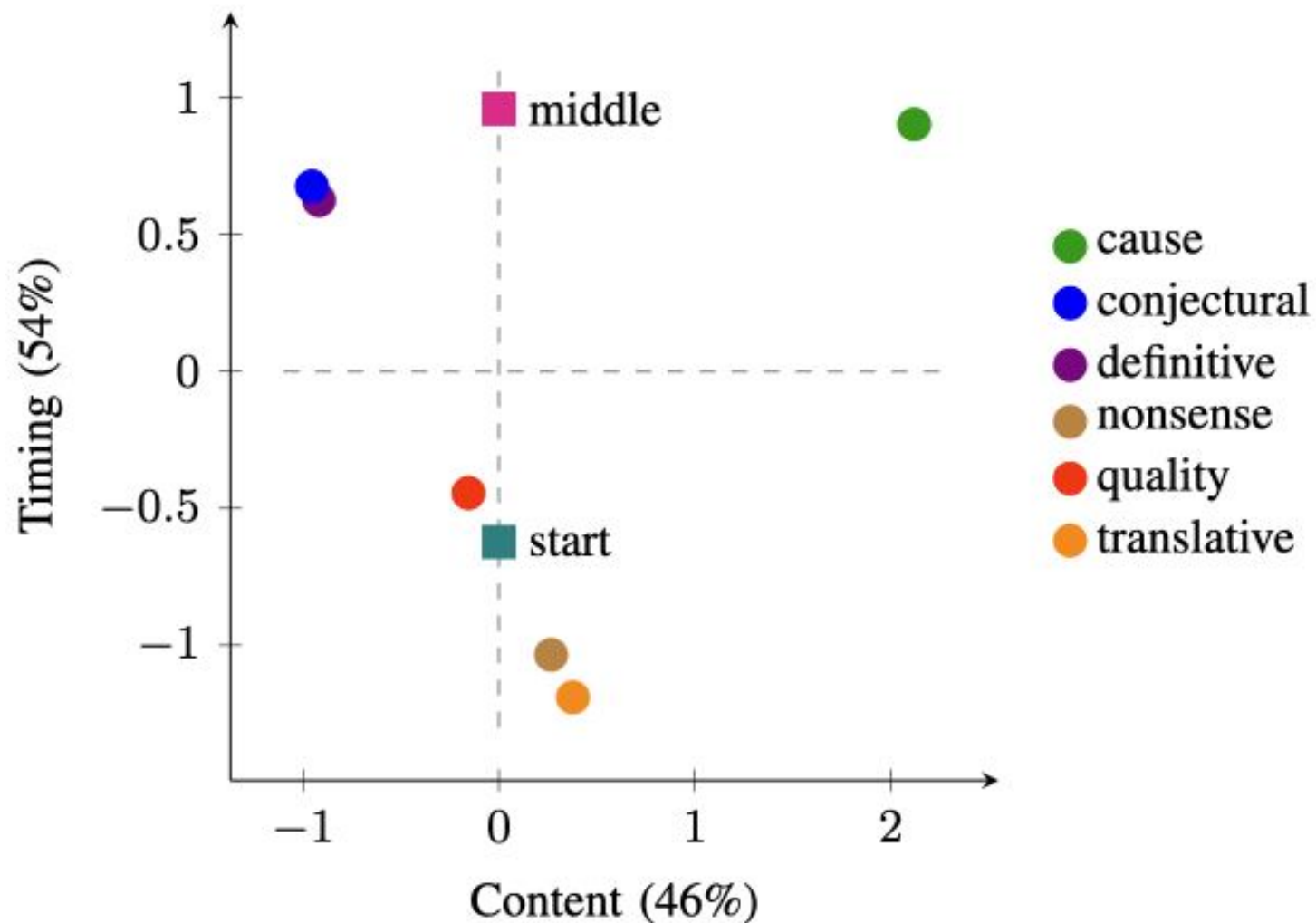
Chi-Square Analysis

$p < 0.001$ | Cramer's V = 0.3 (large effect)

- Temp 0.2 -> 0.9
 - CAUSE, CONJECTURAL, DEFINITIVE, and NONSENSE *nearly doubles* in frequency
 - QUALITY is *halved*
 - TRANSLATIVE decreases slightly

*no significant relationship between temperature and outcome.

Where the LLM is prompted shapes rhetorical direction.



Multiple Correspondence Analysis

$p < 0.001$

Cramer's $V = 0.194$

(small–moderate effect)

- Timing and content explain 54% and 46% of the variance, respectively.
- Axis ranges represent the spread on how strongly they relate to the dimension

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LLM Temperature
0.2 → 0.9

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- *Translative* stays the **same**

Location → Content
p < .05

- *Start: Translative, Quality, Nonsense*
- *Middle: Conjectural & Cause*